ELECTENG 209

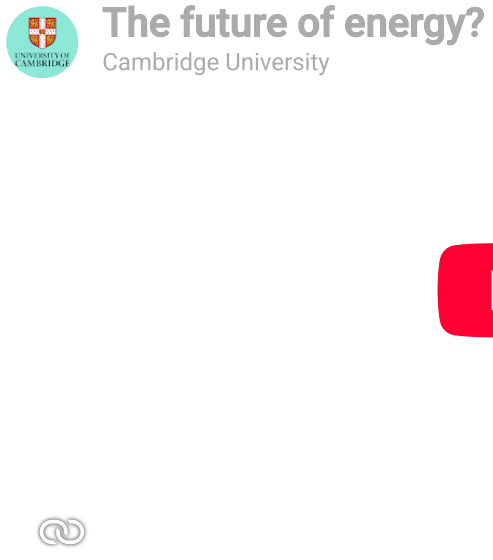
Analog & Embedded Software Design

An Introduction to the Course

Duleepa J Thrimawithana

Background

- High reliance on fossil fuels and associated issues have become increasingly more evident over the last two decades
  - Human-made climate change due to emissions is one of the most alarming concerns

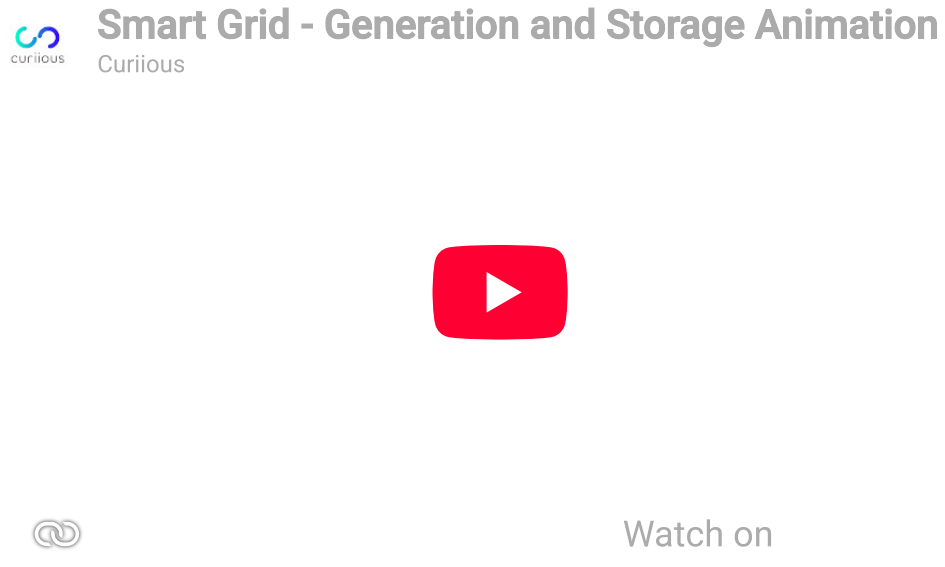


Renewable Energy

- Increasing emphasis on renewable generation, efficient conversion, sustainable utilization and electrification of transportation
  - Many countries have pledged to reduce emissions

Distributed Generation

- Decentralized energy generation (DEG) is gaining popularity due to social, economical and environmental benefits it offers
  - Energy is generated locally at or near point of consumption using micro-scale renewable systems



Energy Management

- Managing energy flow in a DEG systems is very challenging
  - Many distributed sources, loads and storage systems need to be concurrently managed
- A smarter grid with many smart sensors that monitor the flow of energy is a key solution
  - Enables making smart decisions to effectively manage the generation, utilization, storage and retrieval of energy
  - Smart appliances, storage systems and controllers use this information to control the energy flow
- Many smart metering/monitoring solutions have been developed
  - Smart meters report house-hold consumption to retailer
  - Smart energy monitors can monitor energy consumed by a specific load

Eve Energy Monitor
(Taken from evehome.com)

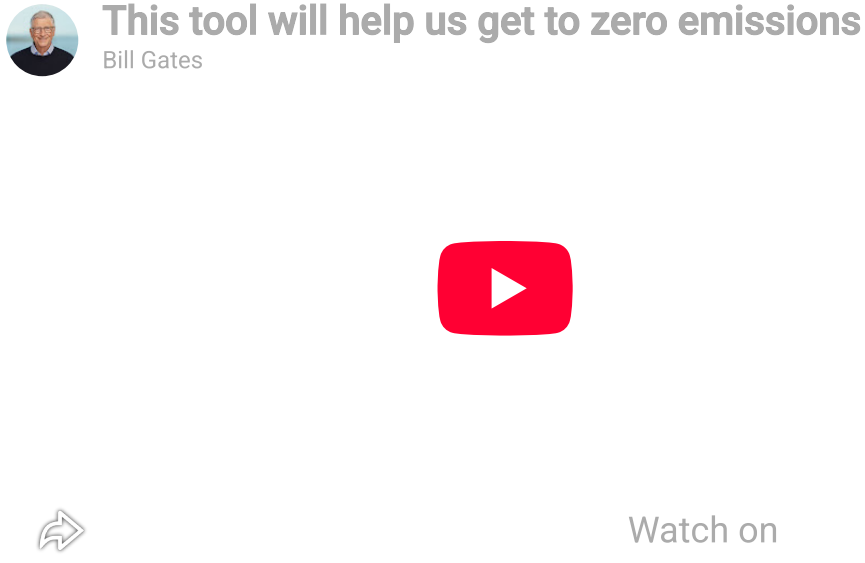
Sense Energy Monitor
(Taken from sense.com)

Renewable Energy Research at UoA

A Few Examples

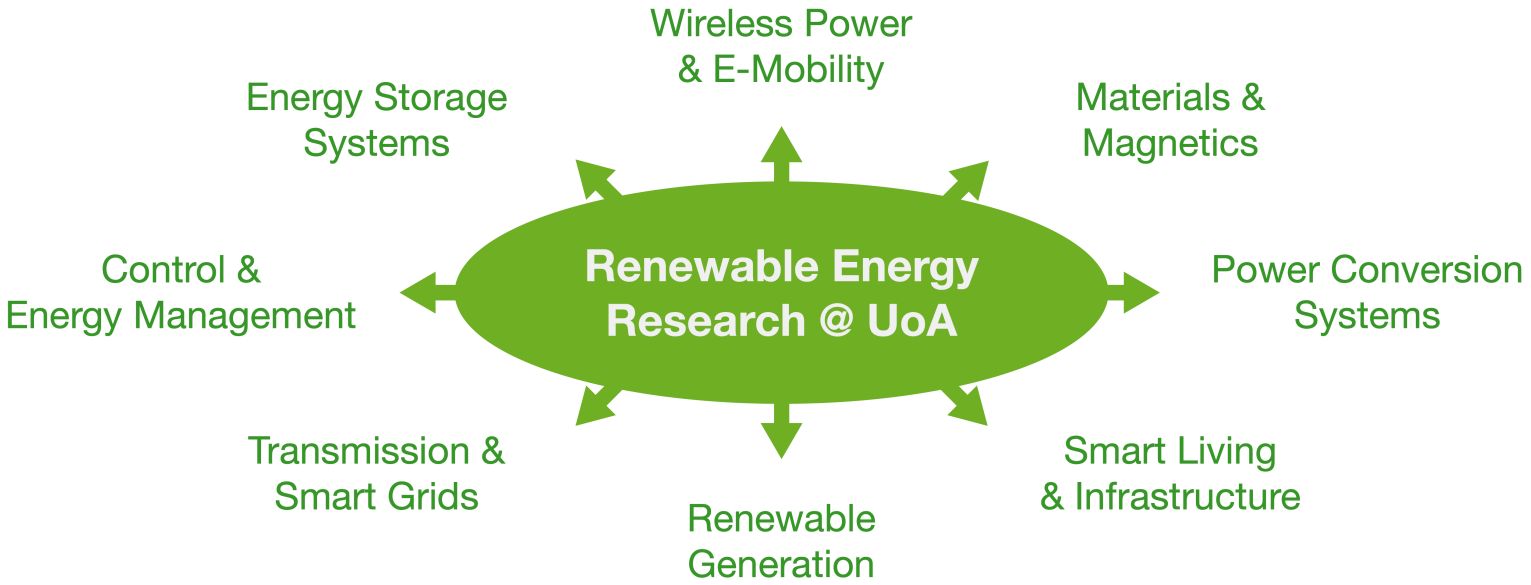
Drive to Achieve Zero Emissions by 2050

- Research on climate change predicts that if we do not achieve **zero emissions** by 2050 the damage to the natural ecosystem will be irreversible
  - Electronics is a key enabling technology that would help achieve a zero emission future



Research Groups

- Across UoA, specially in FoE, there are many research groups working on various renewable energy related research projects
  - Power electronics research group at ECSE is at the forefront of this research



The Power Electronics Research Group

Dist. Prof. Emeritus John Boys

Prof. Grant Covic

Prof. Udaya Madawala

Prof. Patrick Hu

Dr. Duleepa Thrimawithana

35+ Postgraduates

10+ Research Fellows

$100M+ Research Funding

3 Spin-Out Companies

Collaborations with Top US, UK, EU & Asia-Pacific Universities

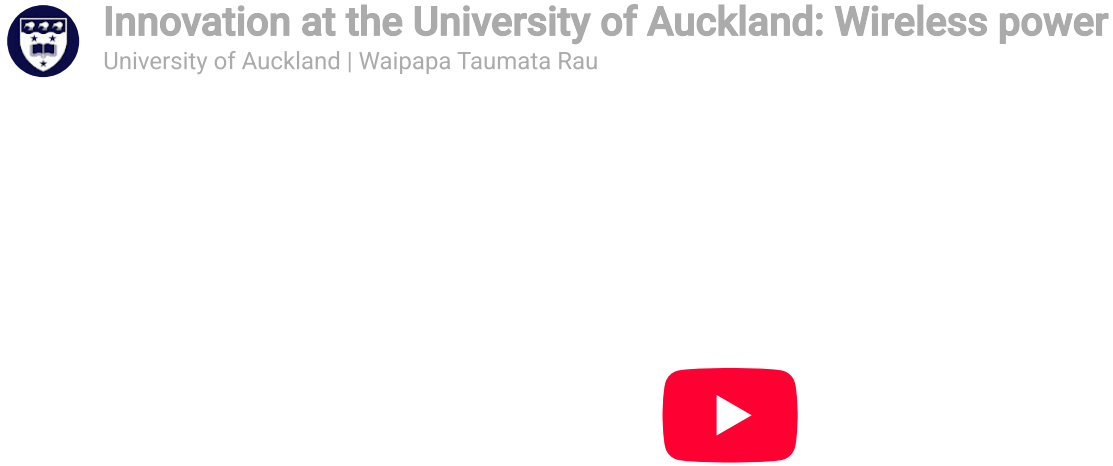
Collaborations with Industry Partners Worldwide

Multidisciplinary Research

Cutting-Edge Laboratory Facilities

100+ Patent Families & 7 Licensees

Pioneers in Wireless Power Transfer (WPT)

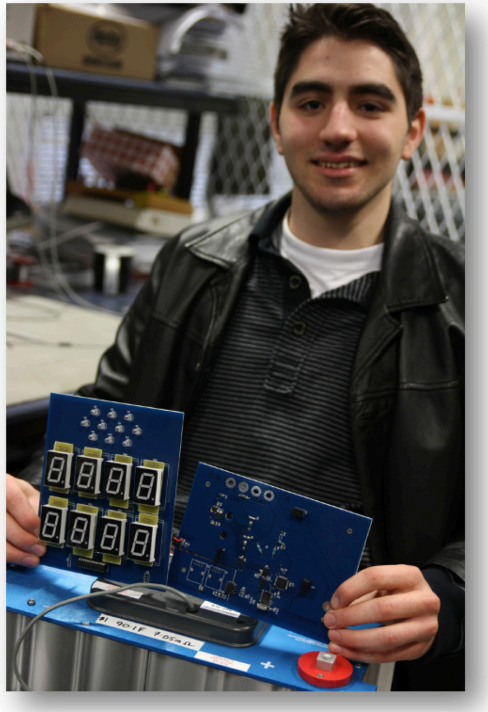


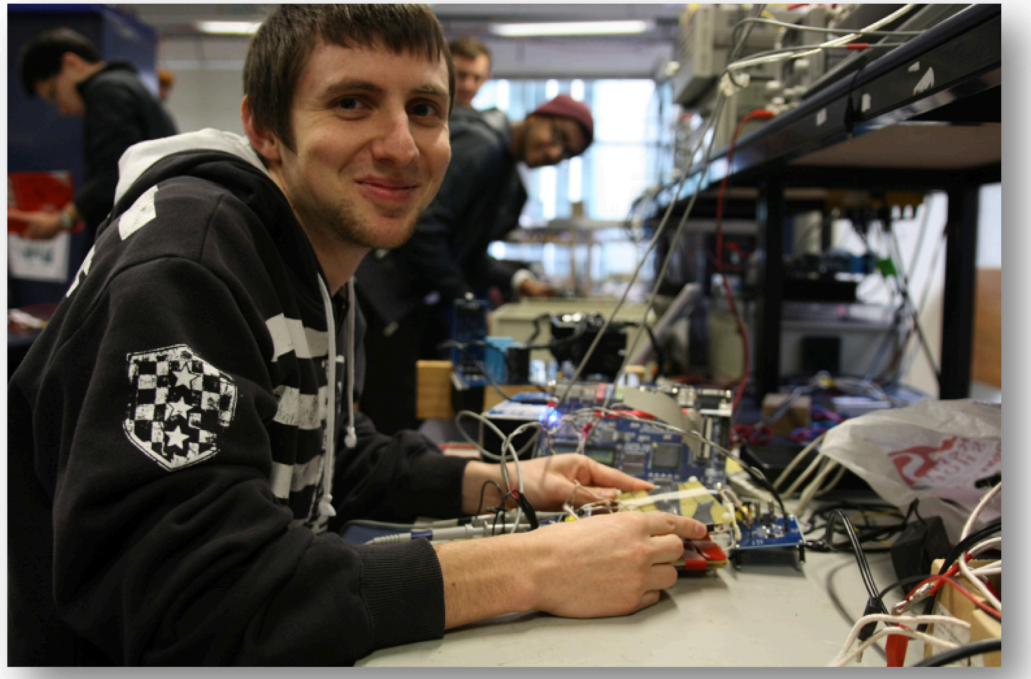
PE Group Connections

- Power electronics research group at ECSE is well connected with top ranking universities as well as industry (a few of these connections are shown below)

PE Research Facilities & Postgraduates

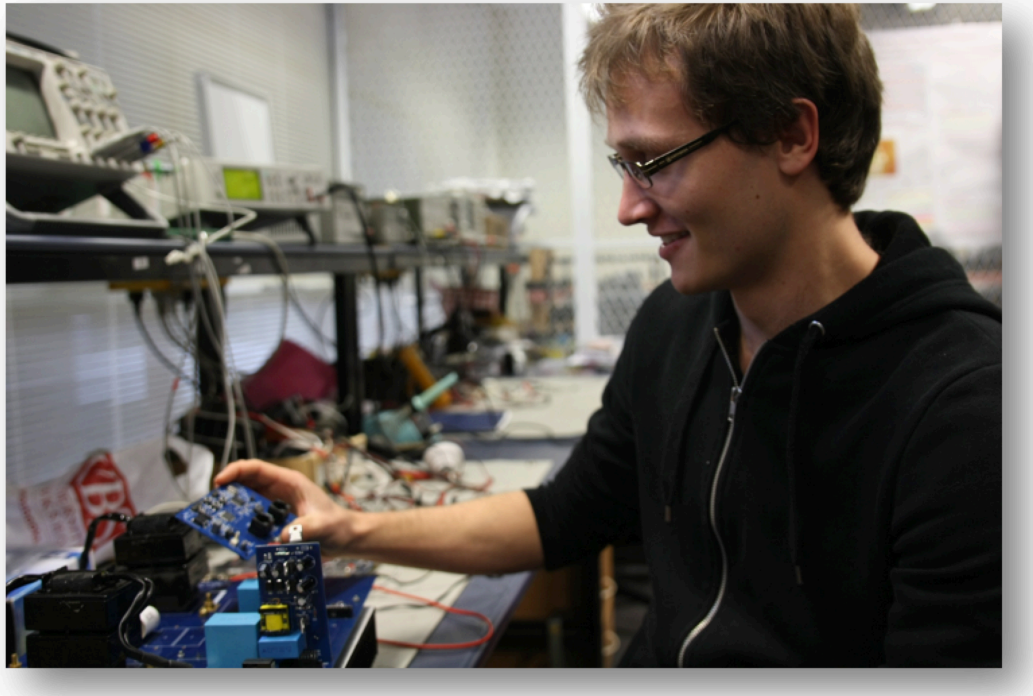
ECSE 2nd Years Undertaking Research

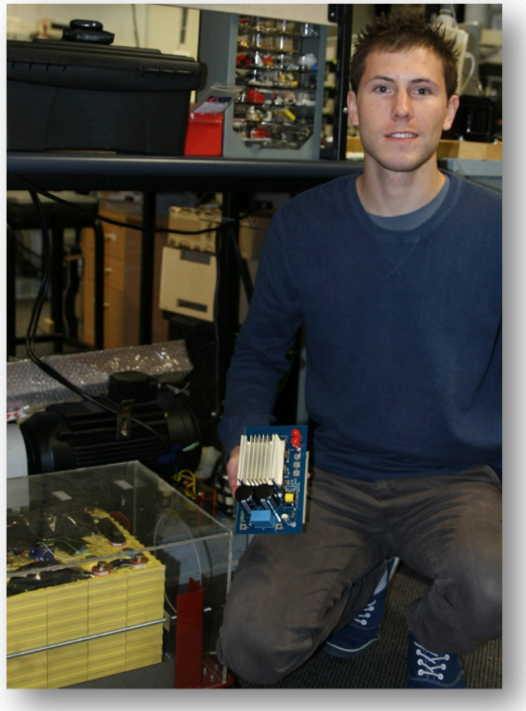




Bashar Dawood (left) & Patrin Illenberger (right) showcasing their summer research projects (2012).
Currently Bashar works as the Managing Director at NGen Automation and Dr. Illenberger works as a hardware engineer at Apple.

ECSE 3rd Years Undertaking Research



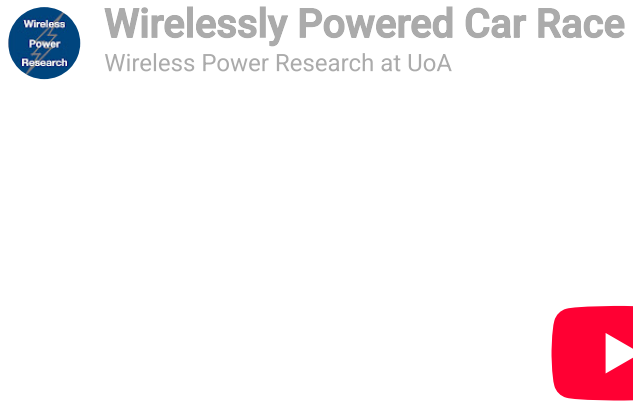


Ryan Kurte (left) & Wynand Malan (right) showcasing their summer research projects (2012).
Currently Ryan works as a R&D engineer at Datamars and Dr. Malan works as a Hardware Development Manager at Apple.

ECSE Final Years Undertaking Research

Hamish O'Neill & Roman Amor showcasing their final year research project (2018).
Currently Hamish works as a hardware engineer at Apple and Roman works as a firmware engineer at Apple.

Intro to Power Electronics (ELECTENG 734)



Students design and develop the power electronics required to wirelessly power a RC car. The best teams compete on the last day of the semester each year to find a winner. This is the final race, in 2018, where the top 2 teams were equally matched.

ELECTENG 734 Prizegiving

Prizegiving (left) & finalists (right) showcasing their final designs (2018).
Sumant, Kunal and a few other engineers from Apple attended the event.

Design of a Smart Energy Monitor

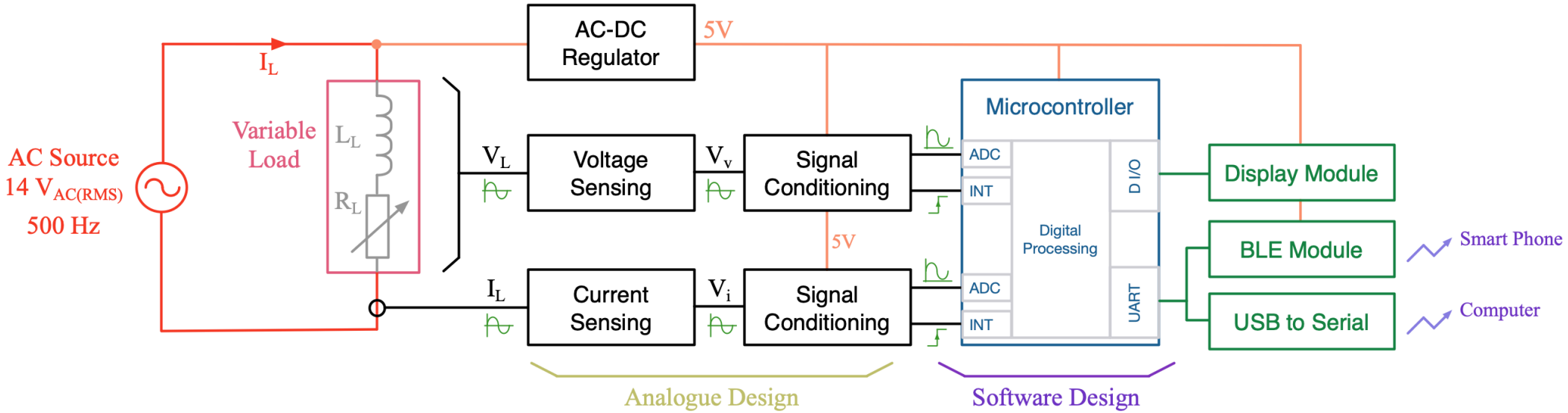
Project Information

The Project

- What is this course about and what should you expect to learn?
  - Gives you an opportunity to put in to practice some of the theories (analog electronics & embedded software development) you learnt to design and engineer a solution to a 'real life problem'
  - During this process you will gain fundamental knowledge, experience, skill-set and professional behavior needed to succeed in the more challenging design projects you will engage in the future
- What would you design and engineer during this course?
  - An energy monitor to measure and display the amount of energy consumed by a household appliance
- How would you achieve this task?
  - The voltage and current at the input to appliance is measured using **analog circuitry** consisting of sensors, amplifiers and filters
  - An **embedded software program** executed on a microcontroller converts the analog signals to digital form and calculates the energy consumption together with other important information
  - The information is shown on a **local LCD display** and also communicated **wirelessly using Bluetooth/WiFi** with other smart devices

System to Implement

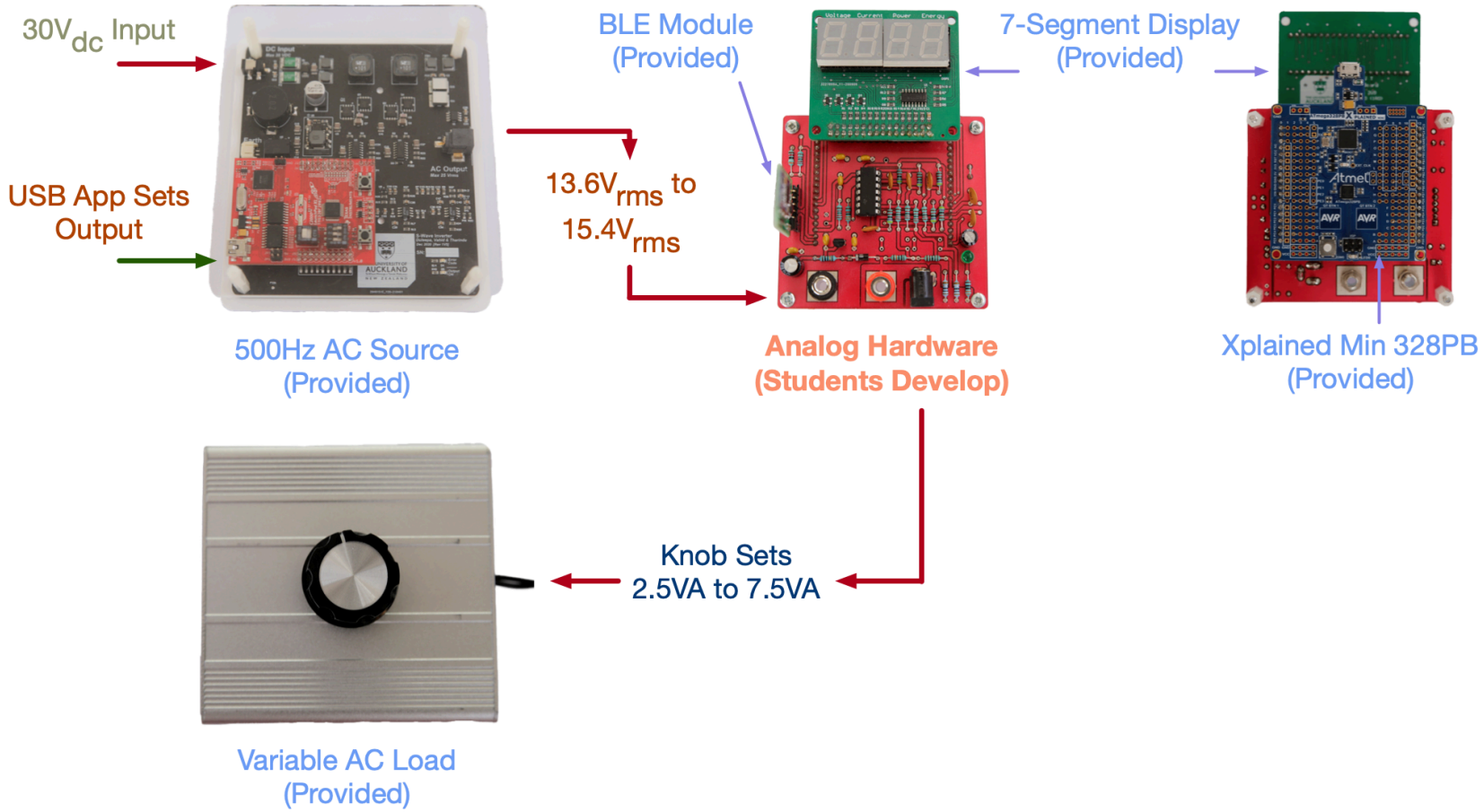
- To simplify the design, we will consider a scaled-down system, which uses a low-voltage AC source
  - The AC source frequency is 500 Hz (oppose to 50 Hz grid frequency) to reduce the size of the AC load
  - An AC load, consisting of a variable resistor in series with a fixed inductor, is used to emulate an house-hold appliance



Key Design Specifications

Parameter	Value
Source Voltage	14 V_{RMS} ± 10%
Source Frequency	500 Hz ± 2%
Load Range	2.5 VA to 7.5 VA
Load Power Factor	0.6 to 0.99
Measurement Accuracy	5% of full-scale reading
ADC Conversion Rate	10 kHz or slower
MCU System Clock	2 MHz
LCD Display Information	Voltage (V_{rms}), Current (mA_{pk}), Power (W)
LCD Scroll Rate	1 s
UART Baud Rate	9600 Baud
Information Transferred Via UART	Voltage, Current, Power and Energy
PCB Size	20000 mm²
PCB Technology	Double Layer with PTH
Device Technology	TH and SMT

Demo of Expected Final Design



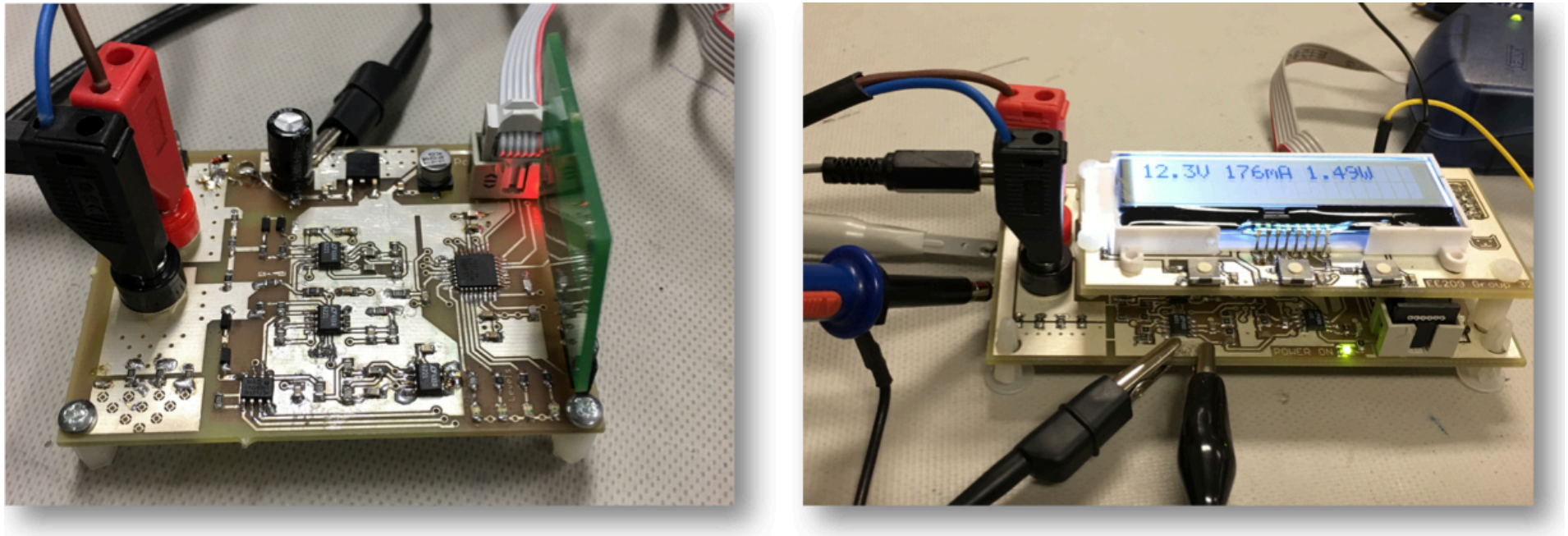
Working in Teams

- You will work in a team of 4
  - You can chose your team members and register your group details on Canvas by the end of 1st week
  - All team members should be from the same lab stream
- Throughout the project work as a team and aim to support each other
  - This is an important part of preparing to be a professional engineer
  - You depend on each other to do well in this course
- Plan your work, document your work and communicate regularly with your team members
  - Use a logbook to plan and document your work
  - Use Slack and Zoom for team communications
  - Use GitHub and G Suite/Office 365 products to collaborate with team members
- Be patient with your team members even if they are not making much progress during the initial stage
  - If things are not working out, after about 1-2 weeks, let us know
  - If a member is not engaging in team work we will ask that person to work alone

Course Calendar

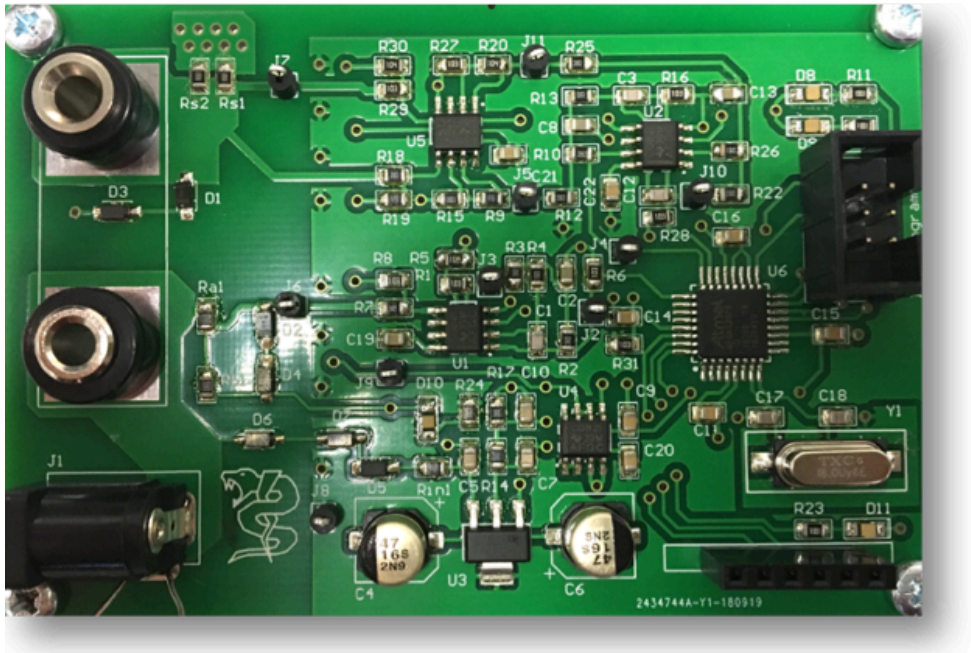
		Monday	Tuesday	Wednesday	Thursday	Friday
Wk 1	Jul	20	21 Lec – Intro *Colab. Tools Wrk*	22 *Colab. Tools Wrk*	23 *AC Circuits Lab (S)*	24 Lec – Sensors *AC Circuits Lab (S)*
Wk 2	Jul	27	26 Lec – Micros *AC Circuits Lab (S)*	29 *AC Circuits Lab (S)*	30 *AC Circuits Lab (M)*	31 Lec – UART *AC Circuits Lab (M)*
Wk 3	Aug	03 *Support*	04 Lec – Dev Tools *Instrument Wrk*	05 *Instrument Wrk*	06 *UART Lab (S)*	07 Lec – Amplifiers *UART Lab (S)*
Wk 4	Aug	10 *Support*	11 Lec – Filters *UART Lab (M)*	12 *UART Lab (M)*	13 *Signal Con. Lab (S)*	14 Lec – Regulators *Signal Con. Lab (S)*
Wk 5	Aug	17 **Test 1**	18 Lec – Proto. & Prac. *Altium Wrk*	19 *Altium Wrk*	20 *Signal Con. Lab (M)*	21 Lec – ADC *Signal Con. Lab (M)*
Wk 6	Aug	24 *Support*	25 Lec – Sig. Pro. **Progress Review**	26 **Progress Review**	27 *ADC Lab (S)*	28 Lec – Timers *ADC Lab (S)*
	Aug/Sep	31	01 *Support*	02	03 *Support*	04
	Sep	07	08 *Support*	09	10 *Support*	11
Wk 7	Sep	14 *PCB Support*	15 Lec – Interrupts PI *ADC Lab (M) + PCBs*	16 *ADC Lab (M) + PCBs*	17 **PCB Submission** *Timers Lab (S)*	18 Lec – Interrupts PII *Timers Lab (S)*
Wk 8	Sep	21 *Support*	22 Lec – Ind. Seminar *Timers Lab (M)*	23 *Timers Lab (M)*	24 *Displays Lab (S)*	25 Lec – Ind. Seminar *Displays Lab (S)*
Wk 9	Sep/Oct	28 *Support*	29 Lec – Review *Displays Lab (M)*	30 *Displays Lab (M)*	01 *Proto. Support*	02 *Proto. Support*
Wk 10	Oct	05 **Test 2**	06 *Proto. Support*	07 *Proto. Support*	08 *Proto. Support*	09 *Proto. Support*
Wk 11	Oct	12 *Support*	13 *Proto. Support*	14 *Proto. Support*	15 *Proto. Support*	16 *Proto. Support*
Wk 12	Oct	19 *Support*	20 **Interviews**	21 **Interviews**	22 **Interviews**	23 **Winners + Pizza**

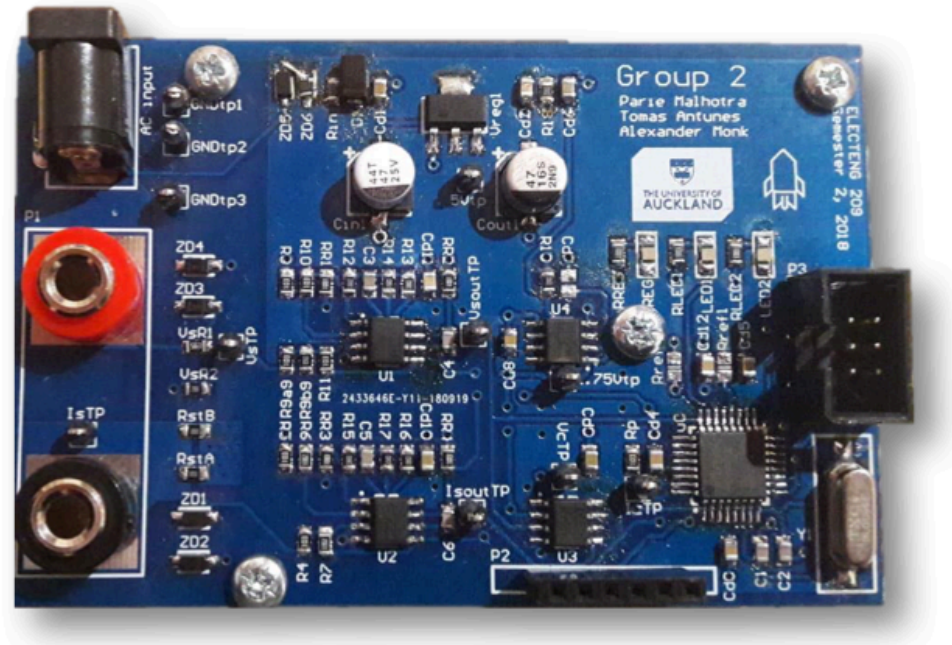
Top Designs from the Past (PI)



SMT designs with a digitally controlled gain stage (left) & a LCD screen (right) developed by teams in 2016.

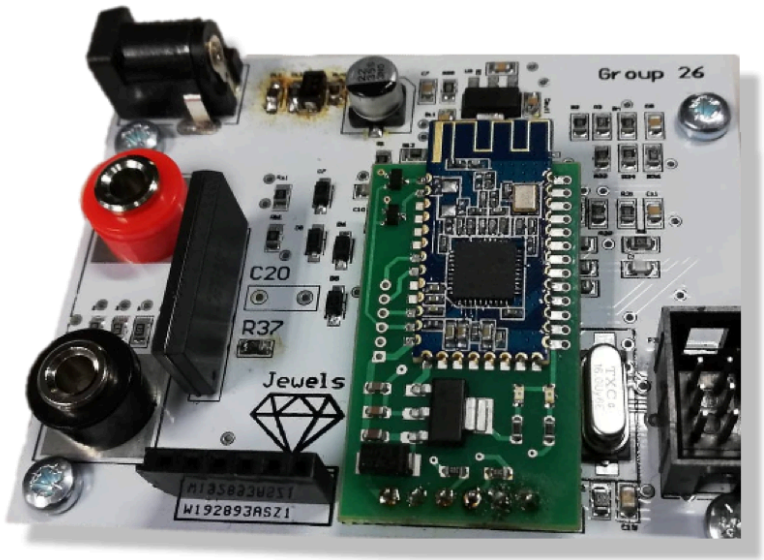
Top Designs from the Past (PII)

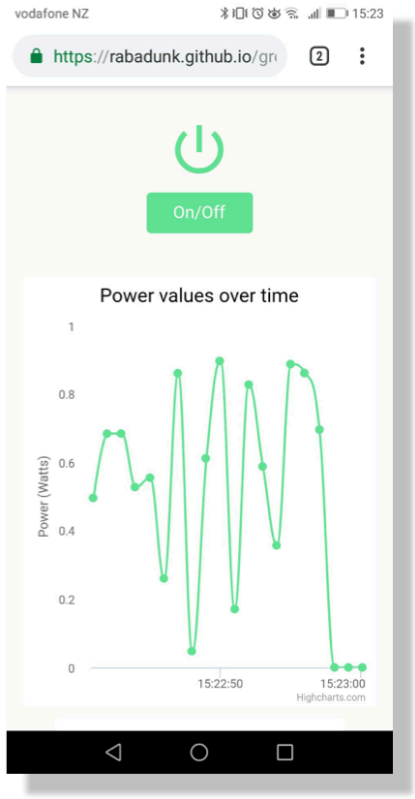




A couple of the SMT designs that were developed by teams in 2018.

Top Designs from the Past (PIII)





A [web application] developed by a team in 2018 to log information.

How to Get an Excellent Grade

- Is the project challenging?
  - Yes, because this is your first real-world engineering design project
  - Due to lack of experience it is common to encounter unforeseen design and integration issues when developing a system, especially with a hardware-software interface
- What are the most challenging aspects?
  - Time management, planing and team work are the key challenges
  - As in a real life project, resources are limited (time, teams' availability, lab space, equipment, software licenses, etc.) and therefore last minute frantic endeavour will not help
- How could I get an excellent grade?
  - Enjoy designing a real-world product and be passionate about your design (this is the key ingredient)
  - Plan and manage your and your team members' time well
  - Throughout the semester you are expected to spend about 12 hours a week on the project
  - Make full use of all lab sessions and staff support provided (there will be in average of about 10 support hours a week)

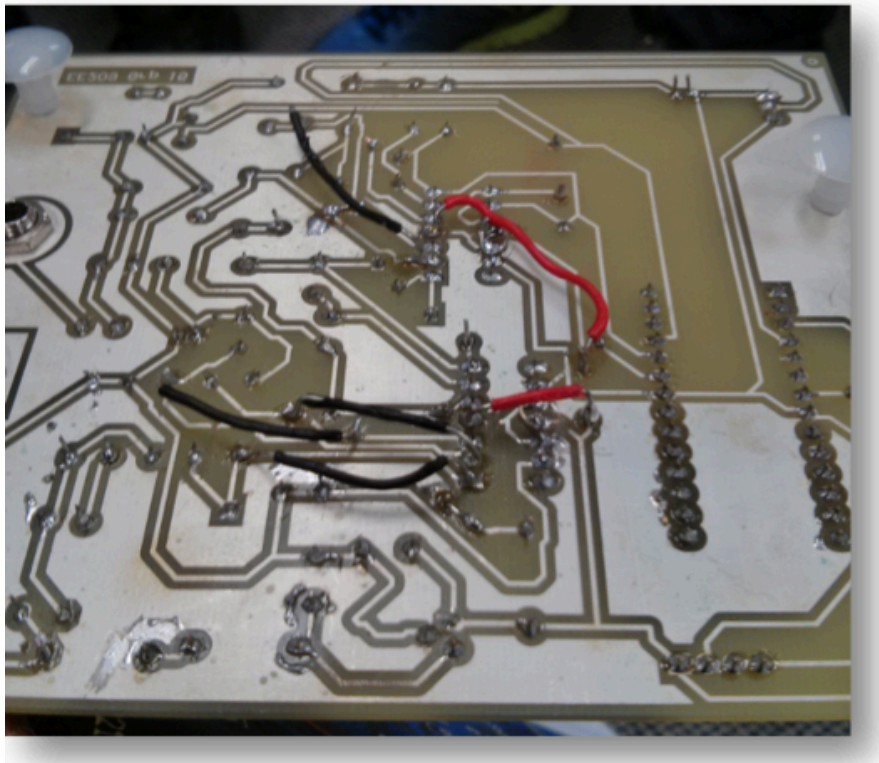
Lab Access

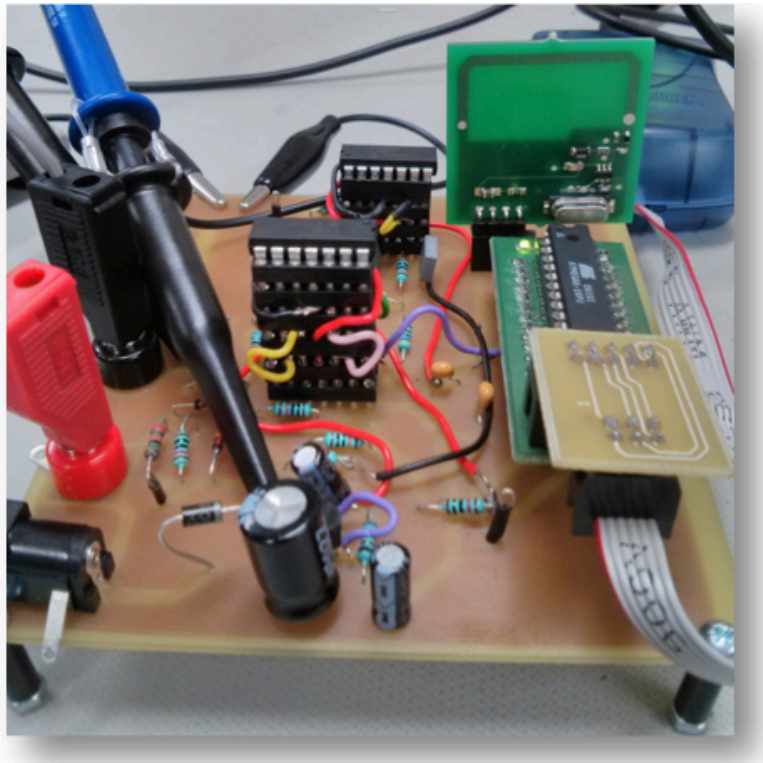
- You will have guaranteed access to the lab only during the scheduled lab sessions
- If you have completed the safety induction, the risk assessment and the other required paper work for after-hours access, you can access the lab anytime between 7.30am and 9pm on weekdays and weekends
  - Make sure to use the labs when not occupied by other courses
- The soldering lab is only open during work hours and it is heavily used by others too
- You are expected to plan your work and use the lab time effectively as it would be difficult to complete the project without access to the resources in the lab
- To help with time management, the labs assessments will guide you through most of the critical tasks
- Note that developing a PCB, assembling and testing it is a time consuming task
  - Allow time during the mid-semester break to complete the PCB design specially if you like to master the PCB design process
- If you are doing the optional design challenges, you must allow time during the mid-semester break to work on this as it takes considerable time
  - There will be limited support available during the mid-semester break

Poor Exemplars - Last Minute Endeavour (PI)

Undergraduate laboratories just a day before the due date - taken by a student at about 11.50 PM (2014).

Poor Exemplars - Last Minute Endeavour (PII)





Consequences of last minute work.

Assessment Components

Details on How to Prepare

Assessment Components

- All assessment components are compulsory
- To obtain a passing grade for this course, **the average score of the 2 tests MUST be over 40%**
  - E.g., if you score 30% in test 1 and 60% in test 2, your average is 45% and you will pass the course
  - E.g., if you score 30% in test 1 and 40% in test 2, your average is 35% and you will **fail the course**
- Although this project will be undertaken in design teams of 4, you will be assessed individually

Component	Weighting
Test 1 & 2 (Written and Simulation Components)	45%
Lecture Quizzes	5%
Weekly Lab Assignments	15%
Midsemester Progress Review	6%
Final Interview	14%
Project Deliverables	15%
Smart Energy Challenge (Optional Bonus)	5%

Tests, Lecture Quizzes & Labs

- The two 22.5% tests will be conducted during weeks 5 and 10 of the semester
  - The written part of the tests will assess your understanding of fundamental concepts
  - The laboratory component of the tests could involve LTspice, coding and instrumentation
- During each of the 15 lecture, we will conduct a 0.5% quizzes using Canvas to evaluate your engagement
  - Quizzes will be available from 1 hour before the lecture to one hour after the lecture
  - They will also be used to track your attendance as the lectures can be thought of *design meetings*
  - The maximum marks you can get will be capped to 5% in case you miss a few lectures due to sickness or other valid reasons
- The 6 lab assignments are designed to guide you through the design and advance your understanding
  - Allow plenty of time, especially if you have forgotten the EE101, EE291 & CS201 contents
  - You can work as a team, but your understanding will be **assessed individually**
  - We will use one of the two weekly lab sessions to help you complete these assignments
  - The remaining weekly lab session will be used to assess your solutions (1%), understanding and engagement (1%) and use of collaboration tools and documentation (0.5%)

Progress Review, Interview & Deliverables

- Just before the mid-semester break we will review your progress
  - An analog circuit that meets all the needs of the project needs to be successfully designed, simulated and validated
  - An embedded software program that can transmit data from the ATmega328P/PB via UART to a terminal should be successfully implemented and validated
  - A short interview will be held to assess your circuit (2%), software (1%) and your understanding (3%)
- During the last week of the semester there will be an interview to evaluate your design as well as your understanding of the subject matter
  - Design will be evaluated based on functionality (5%), measurement accuracy (5%) and quality (5%)
  - The measurement accuracy is a competitive aspect of the project and the team with the best accuracy will get the highest marks
  - Theoretical and practical knowledge of electronics (7%) and programming (7%) concepts you gained during the project will be assessed **individually** during the interview

Smart Energy Challenge & Top PII ECE Design Team

- *Smart Energy Challenge* focuses on improving technology readiness level (TRL) of your design
  - You are expected to implement a prototype using surface mount technology (SMT) components, include the microcontroller onboard plus and preferably package in a small enclosure
  - It should also have wireless communication capabilities (e.g. Bluetooth or WiFi) and an app to display the information on a smart device
- Before attempting the *Smart Energy Challenge* you **must complete all other tasks**
  - You can get up to 4 bonus marks for successfully completing the *Smart Energy Challenge*
- Based on the performance throughout the semester the top 5 design teams will be selected
  - Finalists will get up to 1 bonus marks
- Industry judges from Apple will interview the top 5 teams

PCB Submission

- You are expected to be familiar with the fundamentals of PCB design using Altium Designer
  - PCB design is a key skill you will need to develop as a design engineer
  - Taking a well designed PCB to a job interview will easily help you to stand out from the crowd
- This is the only course in your undergraduate degree where you will be taught how to design a PCB using Altium Designer
  - Use this opportunity to learn the PCB design process and the tool
  - If you like to master the PCB design process, you should aims to spend time during the mid-semester break as we will provide you with additional resources to help you
- Since the primary objective of this course the primary objective of this course is to introduce you to the PCB design process and the tool (Altium Designer), we will be provided with an example PCB design
  - You are allowed to create a replica of this example PCB design

Collaborating with Team Members

- To be a good engineer you also need to develop a number of *soft-skills*
  - Some of the key skills you should aim to develop are management, teamwork and communication
- One of the primary responsibilities you will have as an engineer is to tell other engineers how to build, service, use and update a product you have been involved in designing/developing/building/testing/etc.
  - Thus, engineers are expected to document all details about their work in a logbook
  - The information recorded in a logbook include, design notes, design decisions, calculations, software flow diagrams, schematics, meeting minutes, etc.
- In industry, many different digital tools are used to help with managing teamwork work, communicating with team members and documenting details of the project
  - Slack and GitHub are a few such most commonly used and simple to use tools
- In this project all students are expected to use a logbook together with Slack and GitHub to effectively plan, manage and document the work
  - We will regularly check how you use these tools as part of the assessment process

Resources & Flexible Learning

- Duleepa Thrimawithana (course director), Grant Covic and Patrick Lawton, from the academic staff, will be looking after this course
  - Felix Marattukalam and Kavitha Penneru are your technical facilitators
- Teaching assistance (TAs) will be available to help with lab assignments and the project
  - Bharat Vardani, Tharindu Dharmakeerthi, Fraser McDowell, Kyja McCabe, Yaoyao Zheng, Jensen Benard, Osama Almulla, Charley Shi, Sree Bommineni, James West, Arnesh Sharma, Silva He, Matheus Geronasso, Alastair Charles, Inesh Bhanuka, Leon Lei, Lulu Li, Jeremy Muldoon, Shrirang Ekbote, Damian Robertson, Andre Lee-Joe, Andrew Griffiths, Andy Men, Austin Jensen, Emma Dias, Isabella Wang, Parmjot Singh, Taine Reader, Jasper Ni, Liam Beintken, Junho Bahng, and Boshra Rahnamaee are your TAs
- You could get support in-person during the dedicated lab sessions shown on the lab planner
  - We will also provide support remotely using Ed Discussions, Slack and Zoom
  - You may also request additional staff assisted evening support/tutorial sessions (online or on-campus)

Winter & Respiratory Viruses

- The course will be delivered in person
- There are about 200 students in this course and they are supported by about 30+ TAs and staff
  - Health and safety of everyone is a priority and so, if a TA is ill we would have to find replacements
- We need to try and minimize the spread of respiratory viruses especially since it is winter
  - If you are sick, you can participate virtually in support and assessment sessions or you can request to reschedule your assessments
- At times we may have TA shortages and might be forced to reschedule assessments
  - We will discuss with affected students and organize alternatives
- If you test positive for COVID and think you have had contact with TAs during infectious time period notify Duleepa

Enrolment & Software Installation

- Prerequisite for this course are ELECTENG 291 (or ELECTENG 202) and COMPSYS 201
  - If you have failed any of these courses, please contact us to check if your enrolment is still valid
- Students are also expected to have passed ELECTENG 101
  - ELECTENG 209 heavily relies on the knowledge you should have gained from core ECSE papers (i.e. ELECTENG 101, ELECTENG 202/291 and COMPSYS 201)
  - To help you revise, teaching material from these courses are also made available through the Canvas page for ELECTENG 209
- In some cases, even if you do not meet the prerequisites, after an interview, we may allow you to continue with the course, but we will have to partner you with a suitable group
  - Contact Duleepa for more information
- Follow details on Canvas or the course website to get setup on Ed Discussions, Slack and GitHub
  - Guides detailing how to obtain and install your personal copy of the software we will use in this course (Atmel Studio, Proteus VSM, Altium Designer, LTspice, GitHub Desktop, etc.) are available

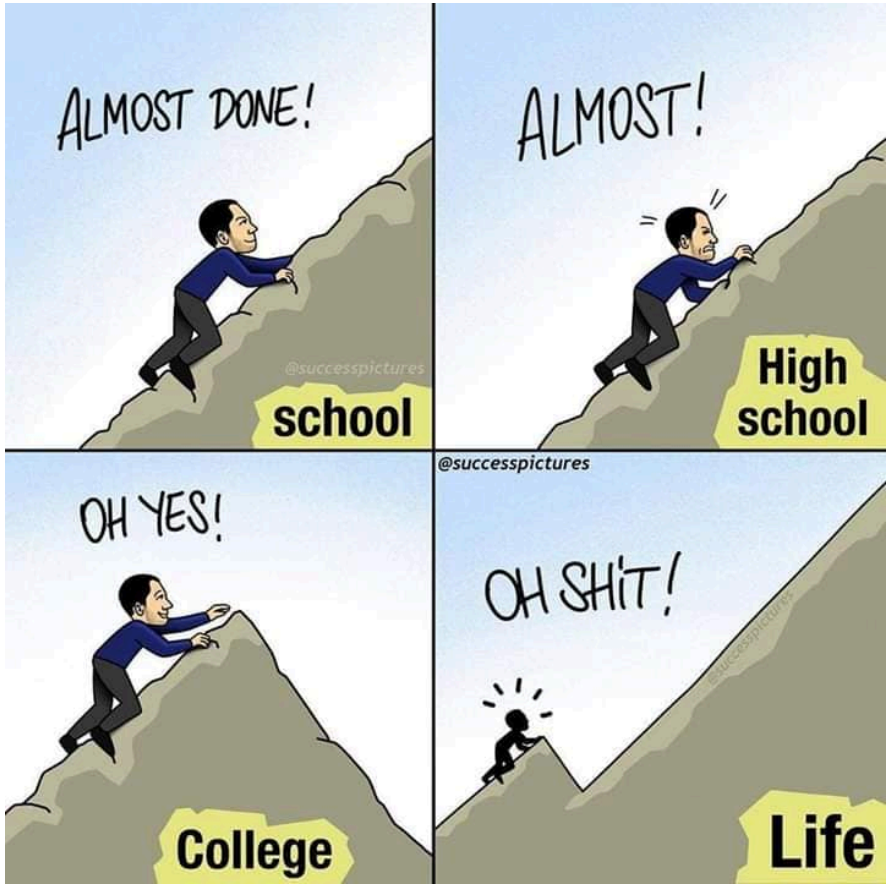
Building a Prototype

- This is a design project and you are expected to develop a hardware prototype
  - This includes breadboarding, assembling PCBs, testing hardware, building final prototype, etc.
- We will provide resources and tools to get started on the project before you get to the stage of building your own prototype analog circuitry (using the provided partially assembled PCB)
  - You can use Proteus VSM and the Xplained Mini development boards to develop and validate most of your software
  - We will provide a hardware emulator that can be connected with the Xplained Mini for further debugging and validation by emulating the signals you expect to get from your own analog circuitry
- The lab benches will be equipped with low-voltage AC power sources and variable load banks
  - The sources generate 12.6 V_{RMS} to 15.4 V_{RMS}, 500 Hz AC voltage
  - The variable resistor in the load can be used to change the VA drawn by it between 2.5VA and 7.5VA
- We will have a number of workshops to teach you how to build and test your hardware

AI Chatbots

- In this course we encourage you to **RESPONSIBLY** use AI chatbots like ChatGPT to help improve your understanding
  - Can be very useful for the firmware component
- Primary use of these tools should be to help improve your understanding
  - **DO NOT** use these tools as a shortcut to bypass learning
- Remember that more than 70% of your marks are individually assessed based on your understanding and design skills
  - More than 20% comes from answering technical questions asked by assessors
  - More than 20% comes from completing practical components of the tests
- Thus to pass this course you need to really master your understanding and skills

Looking Ahead



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